

EXP:18

Exp No 18: Determination and plotting of Antenna Half power beamwidth.

Aim: To determine and plot Half Power Beamwidth (HPBW) and to determine the angular width at -3 dB points

```
=====
"      ANTENNA HPBW RESULT      "
=====
"Number of elements n = "
6.
"First -3 dB point = "
81.5
"Second -3 dB point = "
98.5
"HPBW in degrees = "
17.
-->
```

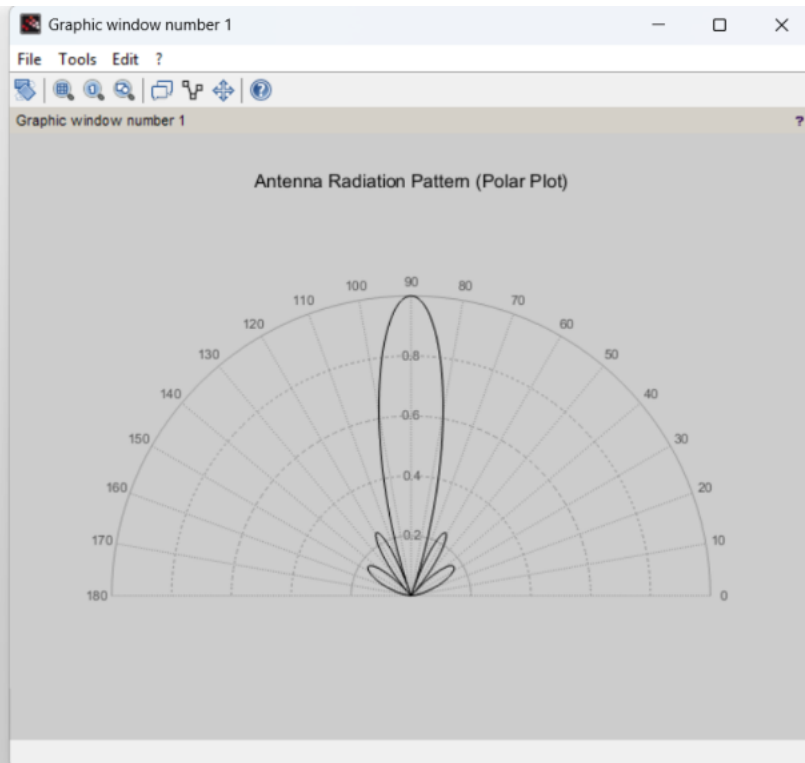
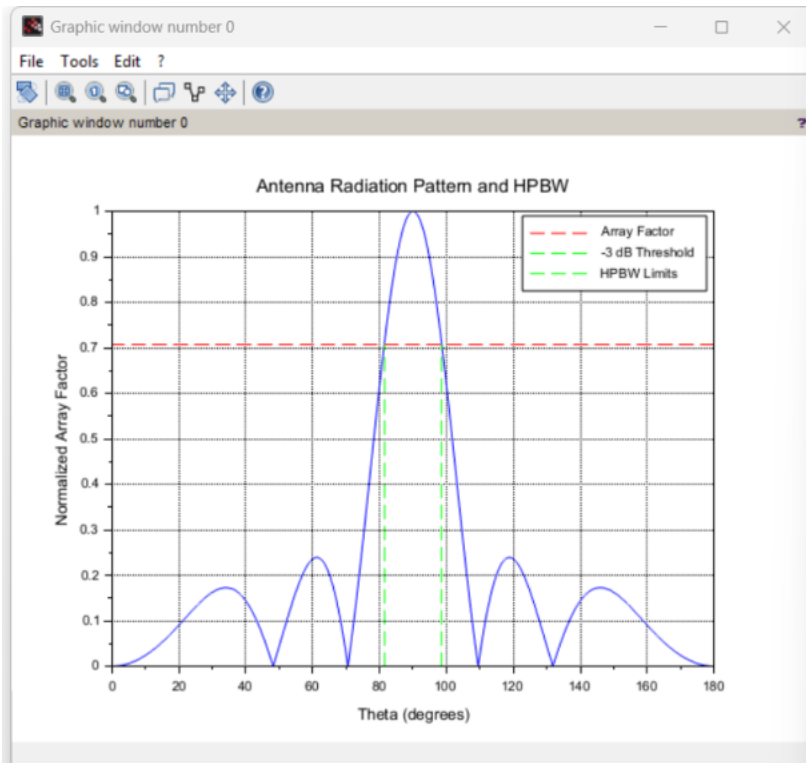
```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X
1  clc;
2  clear;
3  close;
4  n = 6;
5  theta = 0:0.1:180;
6  theta_rad = theta * %pi / 180;
7  d = 0.5;
8  AF = zeros(1, length(theta));
9  for k = 1:length(theta)
10     psi = 2 * %pi * d * cos(theta_rad(k));
11     if abs(sin(psi/2)) < 1e-6 then
12         AF(k) = 1;
13     else
14         AF(k) = abs(sin(n*psi/2) / ...
15             (n*sin(psi/2)));
16     end
17 end
18 AF = AF / max(AF);
19 threshold = 1 / sqrt(2);
20 indices = find(AF >= threshold);
21 theta1 = theta(indices(1));
22 theta2 = theta(indices(6));
23 HPBW = theta2 - theta1;
24 disp("=====");
25 disp(".....ANTENNA HPBW RESULT");
26 disp("=====");
27 disp("Number of elements n = ");
28 disp(n);
29 disp("First -3 dB point = ");
30 disp(theta1);
31 disp("Second -3 dB point = ");
32 disp(theta2);
33 disp("HPBW in degrees = ");
34 disp(HPBW);
35 clf();
36 plot(theta, AF);
37 plot(theta, threshold * ones(1, length(theta)), "r--");
38 plot([theta1 theta1], [0 threshold], "g--");
39 plot([theta2 theta2], [0 threshold], "g--");
40 xlabel("Theta (degrees)");
41 ylabel("Normalized Array Factor");
```

```

41 ylabel("Normalized-Array-Factor");
42 title("Antenna-Radiation-Pattern-and-HPBW");
43 legend(["Array-Factor"; "-3-dB-Threshold"; "HPBW-Limits"]);
44 xgrid();
45 figure();
46 polarplot(theta_rad, AF);
47 title("Antenna-Radiation-Pattern-(Polar-Plot)");

```

OUTPUT:



Case Study 1: Beamwidth Comparison for Different Directivity Factors

```

=====
"      HPBW COMPARISON"
=====
"For n = 4:"
"Half Power Beamwidth = "
65.530199
"For n = 6:"
"Half Power Beamwidth = "
54.027205
"For n = 8:"
"Half Power Beamwidth = "
47.016100
-->

1  clc;
2  clear;
3  close;
4  theta = -90:0.1:90;
5  theta_rad = theta * %pi / 180;
6  n1 = 4;
7  n2 = 6;
8  n3 = 8;
9  G4 = (cos(theta_rad)).^n1;
10 G6 = (cos(theta_rad)).^n2;
11 G8 = (cos(theta_rad)).^n3;
12 threshold = 0.5;
13 theta4 = acos((0.5)^(1/n1)) * 180 / %pi;
14 theta6 = acos((0.5)^(1/n2)) * 180 / %pi;
15 theta8 = acos((0.5)^(1/n3)) * 180 / %pi;
16 HPBW4 = 2 * theta4;
17 HPBW6 = 2 * theta6;
18 HPBW8 = 2 * theta8;
19 disp("=====");
20 disp(".....HPBW-COMPARISON");
21 disp("=====");
22 disp("For n = 4:");
23 disp("Half-Power-Beamwidth = ");
24 disp(HPBW4);
25 disp("For n = 6:");
26 disp("Half-Power-Beamwidth = ");
27 disp(HPBW6);
28 disp("For n = 8:");
29 disp("Half-Power-Beamwidth = ");
30 disp(HPBW8);
31 clf();
32 plot(theta, G4);
33 plot(theta, G6);
34 plot(theta, G8);
35 plot(theta, threshold * ones(1, length(theta)), "r--");
36 plot([-theta4 theta4], [threshold threshold], "g--");
37 plot([-theta6 theta6], [threshold threshold], "b--");
38 plot([-theta8 theta8], [threshold threshold], "m--");
39 xlabel("Theta (degrees)");
40 ylabel("Normalized Gain");
41 title("Radiation Pattern and HPBW Comparison");

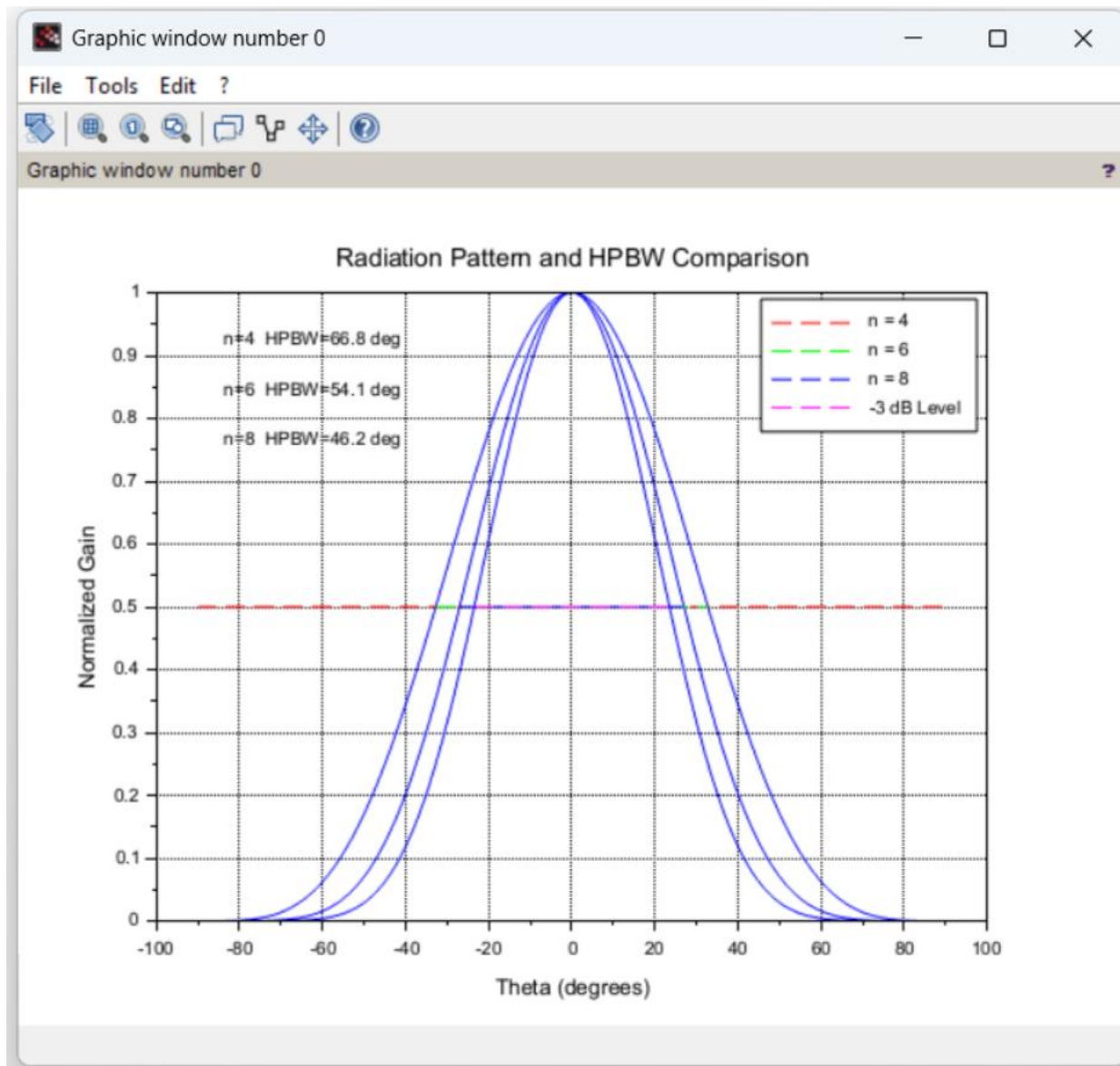
```

```

42 legend(["n = 4"; "...",
43         "n = 6"; "...",
44         "n = 8"; "...",
45         "-3-dB-Level"]);
46 xgrid();
47 xstring(-85, -0.90, "n=4--HPBW=66.8-deg");
48 xstring(-85, -0.82, "n=6--HPBW=54.1-deg");
49 xstring(-85, -0.74, "n=8--HPBW=46.2-deg");
50

```

OUTPUT:



Case Study 2: Effect of Frequency on HPBW

```

=====
"  EFFECT OF FREQUENCY ON HPBW"
=====
"For frequency = 1 GHz"
"Wavelength (m) = "
0.3
"HPBW (degrees) = "
21.
"For frequency = 3 GHz"
"Wavelength (m) = "
0.1
"HPBW (degrees) = "
7.
"For frequency = 6 GHz"
"Wavelength (m) = "
0.05
"HPBW (degrees) = "
3.5

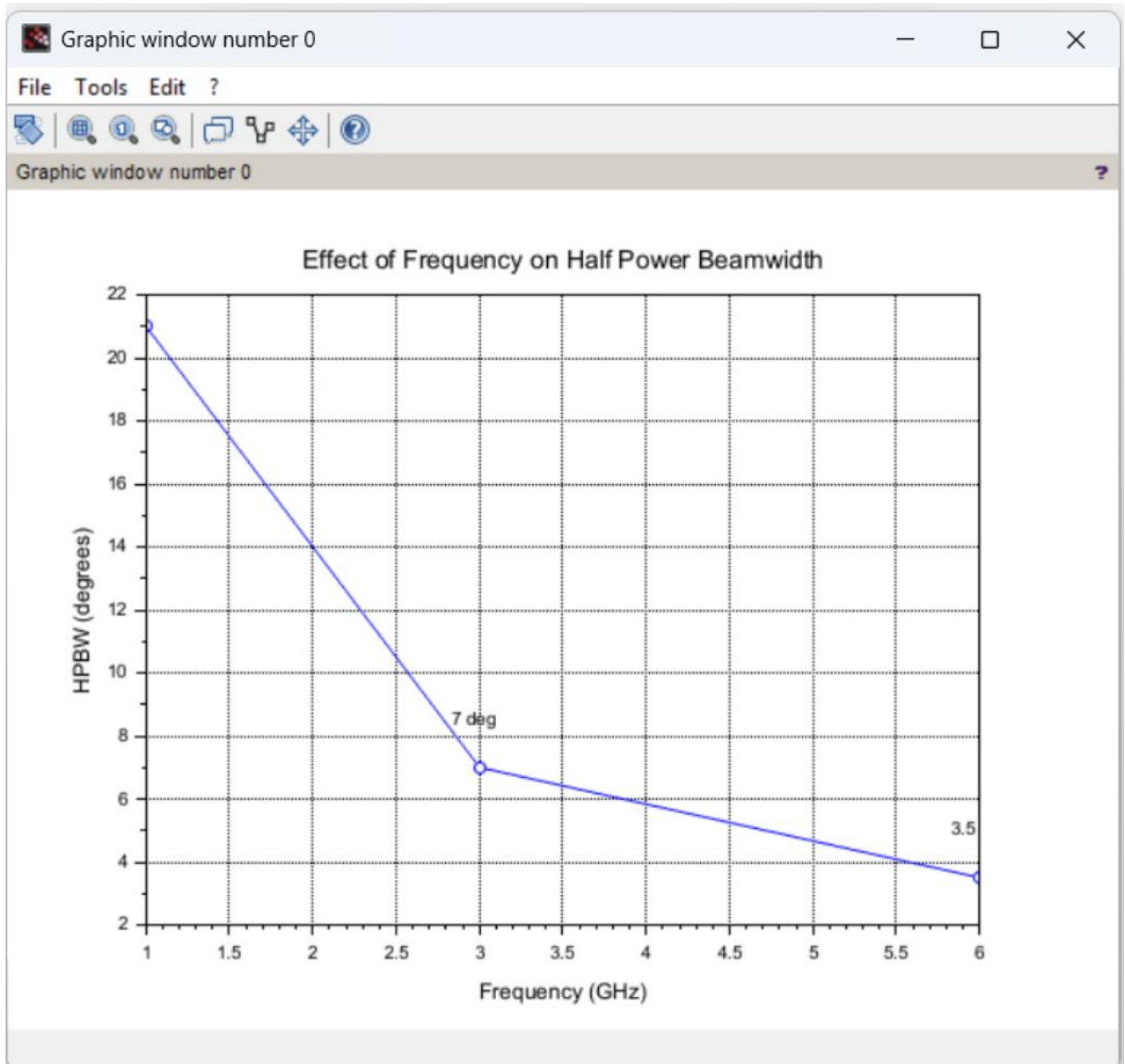
```

```

*exp333.sci X
3 close;
4 c = 3e8; .....//Speed of light (m/s)
5 D = 1; .....//Diameter of parabolic reflector (m)
6 eta = 0.9; .....//Antenna efficiency
7 f1 = 1e9; .....//1.GHz
8 f2 = 3e9; .....//3.GHz
9 f3 = 6e9; .....//6.GHz
10 lambda1 = c / f1;
11 lambda2 = c / f2;
12 lambda3 = c / f3;
13 HPBW1 = 70.* lambda1;
14 HPBW2 = 70.* lambda2;
15 HPBW3 = 70.* lambda3;
16 disp("=====");
17 disp("...EFFECT OF FREQUENCY ON HPBW");
18 disp("=====");
19 disp("For frequency = 1.GHz");
20 disp("Wavelength (m) = ");
21 disp(lambda1);
22 disp("HPBW (degrees) = ");
23 disp(HPBW1);
24 disp("For frequency = 3.GHz");
25 disp("Wavelength (m) = ");
26 disp(lambda2);
27 disp("HPBW (degrees) = ");
28 disp(HPBW2);
29 disp("For frequency = 6.GHz");
30 disp("Wavelength (m) = ");
31 disp(lambda3);
32 disp("HPBW (degrees) = ");
33 disp(HPBW3);
34 frequency = [1 3 6];
35 HPBW = [HPBW1 HPBW2 HPBW3];
36 clf();
37 plot(frequency, HPBW, 'bo-');
38 xlabel("Frequency (GHz)");
39 ylabel("HPBW (degrees)");
40 title("Effect of Frequency on Half-Power Beamwidth");
41 xgrid();
42 xstring(0.8, HPBW1+1, "21-deg");
43 xstring(2.8, HPBW2+1, "7-deg");
44 xstring(5.8, HPBW3+1, "3.5-deg");

```

OUTPUT:

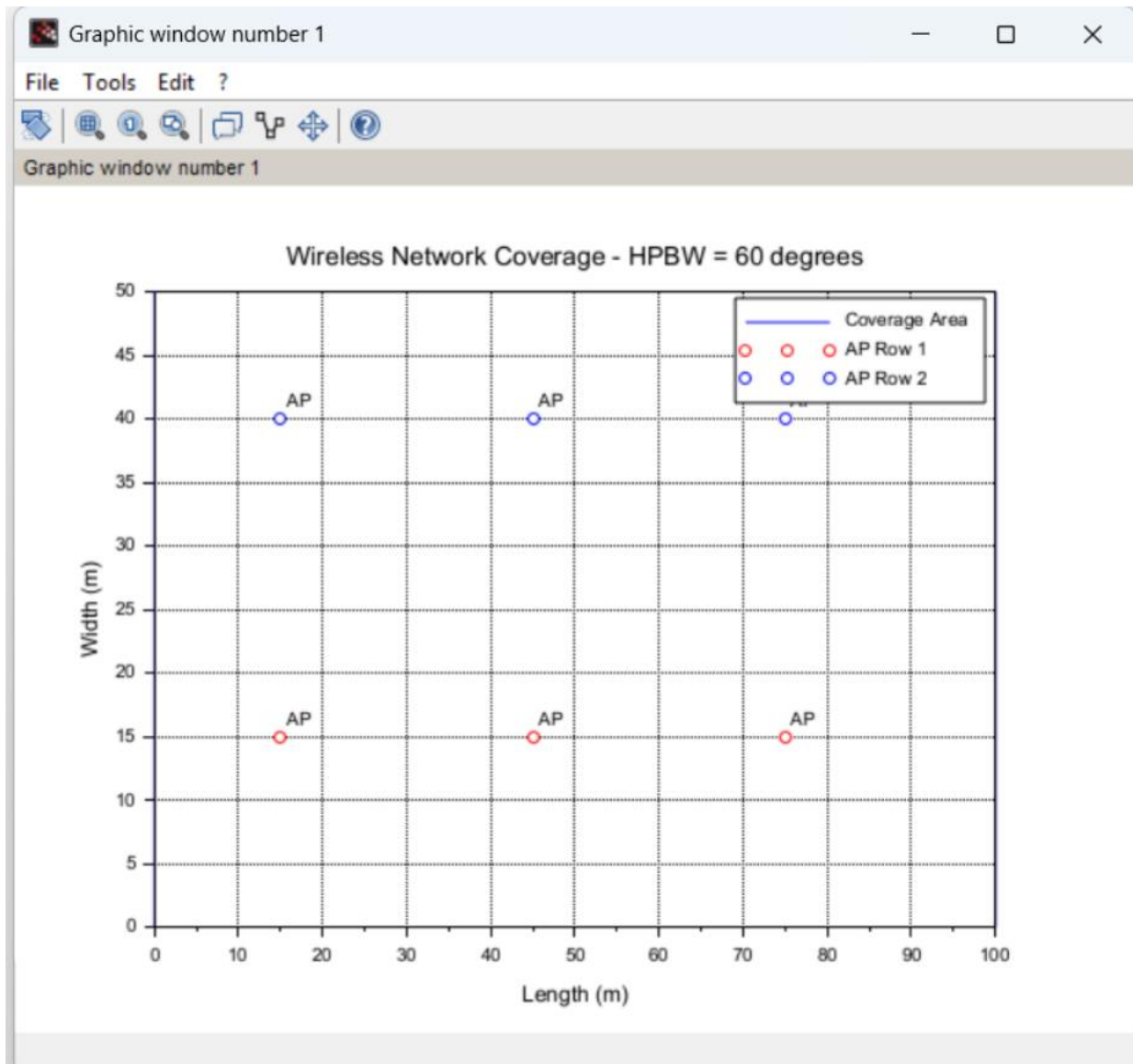


Case Study 3: Practical Implementation of HPBW in Wireless Networks

```
"=====
" PRACTICAL IMPLEMENTATION OF HPBW
"=====
Area = 100 m x 50 m
HPBW = 60 degrees
Range = 30 m
Total Area = 5000.00 m^2
Coverage Area of one AP = 471.24 m^2
Number of APs required = 11
"=====
-->
```

```
*exp333.sci X
1 clc;
2 clear;
3 L = 100;
4 W = 50;
5 HPBW = 60;
6 r = 30;
7 TotalArea = L * W;
8 CoverageArea = (HPBW / 360) * pi * r^2;
9 N = ceil(TotalArea / CoverageArea);
10 disp("=====");
11 disp("PRACTICAL IMPLEMENTATION OF HPBW");
12 disp("=====");
13 mprintf("Area = %d m x %d m\n", L, W);
14 mprintf("HPBW = %d degrees\n", HPBW);
15 mprintf("Range = %d m\n", r);
16 mprintf("Total Area = %.2f m^2\n", TotalArea);
17 mprintf("Coverage Area of one AP = %.2f m^2\n", CoverageArea);
18 mprintf("Number of APs required = %d\n", N);
19 disp("=====");
20 APx = [15 45 75];
21 APy = [15 15 15];
22 APx2 = [15 45 75];
23 APy2 = [40 40 40];
24 scf(1);
25 clf();
26 plot([0 100 100 0 0], [0 0 50 50 0]);
27 plot(APx, APy, "ro");
28 plot(APx2, APy2, "bo");
29 for i = 1:3
30     xstring(APx(i), APy(i), "-AP");
31 end
32 for i = 1:3
33     xstring(APx2(i), APy2(i), "-AP");
34 end
35 xlabel("Length (m)");
36 ylabel("Width (m)");
37 title("Wireless Network Coverage - HPBW = 60 degrees");
38 legend(["Coverage Area"; "AP-Row-1"; "AP-Row-2"]);
39 xgrid();
```

OUTPUT:



Case Study 4: Validation of HPBW with Measured Data

```

=====
" VALIDATION OF HPBW WITH MEASURED DATA"
=====
Frequency = 2.4 GHz
Patch Length = 0.05 m
Wavelength = 0.125 m
" "
Theoretical HPBW = 0.31 degrees
Experimental -3 dB Point 1 = -44.00 degrees
Experimental -3 dB Point 2 = 44.00 degrees
Experimental HPBW = 88.00 degrees
Difference = 87.69 degrees
=====
-->

```

```

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1  clc;
2  clear;
3  close;
4  c = 3e8; % Speed of light
5  f = 2.4e9; % Frequency = 2.4 GHz
6  L = 0.05; % Patch length = 5 cm
7  lambda = c / f;
8  HPBW_theory = lambda * L * 50;
9  theta = -90:1:90;
10 P = cos(theta * pi / 180).^2;
11 P = P / max(P);
12 PdB = 10 * log10(P + 0.000001);
13 threshold = -3;
14 indices = find(PdB >= threshold);
15 theta1 = theta(indices(1));
16 theta2 = theta(indices(end));
17 HPBW_exp = theta2 - theta1;
18 difference = abs(HPBW_exp - HPBW_theory);
19 disp('=====');
20 disp(' VALIDATION OF HPBW WITH MEASURED DATA');
21 disp('=====');
22 fprintf('Frequency = %.1f GHz\n', f/1e9);
23 fprintf('Patch Length = %.2f m\n', L);
24 fprintf('Wavelength = %.3f m\n', lambda);
25 disp('.');
26 fprintf('Theoretical HPBW = %.2f degrees\n', HPBW_theory);
27 fprintf('Experimental -3 dB Point 1 = %.2f degrees\n', theta1);
28 fprintf('Experimental -3 dB Point 2 = %.2f degrees\n', theta2);
29 fprintf('Experimental HPBW = %.2f degrees\n', HPBW_exp);
30 fprintf('Difference = %.2f degrees\n', difference);
31 disp('=====');
32 scf(1);
33 clf();
34 plot(theta, PdB);
35 plot(theta, -3 * ones(1, length(theta)), 'r--');
36 plot([theta1 theta1], [-40 -3], 'g--');
37 plot([theta2 theta2], [-40 -3], 'g--');
38 xlabel('Angle (degrees)');
39 ylabel('Normalized Power (dB)');
40 title('Microstrip Patch Antenna Radiation Pattern - 2.4 GHz');
41 legend(['Measured Radiation Pattern'; ...

```

```

42 ..... "-3 dB Level"; ...
43 ..... "HPBW Limits"]);
44 xgrid();
45

```

OUTPUT:

